

Gebran Kastoun

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EDUCATION

Cornell University

Ithaca, NY

BS in Electrical and Computer Engineering, Computer Science (GPA: 3.64/4.0)

May 2028

- **Relevant Coursework:** Power Electronics, Embedded Systems, Microelectronics, Circuits, Computer Architecture, Digital System Design with Microcontrollers, Systems Programming

EXPERIENCE

SpaceX

May 2026 – Aug. 2026

Avionics Hardware Intern

Hawthorne, CA

- Designed two 4-layer PCB interface cards for flight avionics, implementing point-of-load power regulation and differential signal interfaces to integrate replacement flight hardware at 80% lower cost
- Served as Responsible Engineer for a flight-critical reentry-vehicle avionics box, owning hardware from requirements through integration and resolving electrical, mechanical, manufacturing, and software interface issues
- Bench-tested modified power-conversion hardware and updated two avionics units for revised vehicle power requirements, performing electrical verification and supporting system integration
- Investigated a hardware failure by reviewing PCB layout, isolating 6 candidate failure points, and defining targeted bench tests to drive root-cause analysis

Cornell CUAir – 2nd Place, 2025 C-UASC Competition

Nov. 2024 – Present

Electrical Subteam Lead & Aircraft Pilot

Ithaca, NY

- Designed and brought up an ATmega328P gimbal stabilization PCB integrating IMU sensing, servo actuation, UART communication, programming, and debug interfaces, reducing system size and weight by 50% and delivering CUAir's first functional gimbal in 6 years
- Developed C++ firmware for closed-loop stabilization using I²C IMU feedback, PID control, and PWM servo actuation while streaming live angle telemetry over UART through a Raspberry Pi to the ground station
- Debugged power, I²C, UART, and PWM interfaces during board bring-up using oscilloscopes and multimeters; incorporated LEDs, jumper pads, and test points to accelerate system integration and troubleshooting
- Own aircraft electrical and avionics architecture, leading design reviews and hardware/software integration across 9 projects while mentoring 12 members and coordinating electrical, mechanical, and software interfaces during test flights

Cornell University

Jan. 2026 – Present

Teaching Assistant

Ithaca, NY

- **ECE 4760 – Digital Systems Design Using Microcontrollers:** Support 80 undergraduate and MEng students debugging RP2350 systems in C across peripherals, real-time firmware, concurrency, and hardware/software integration
- **ECE 2100 – Circuits for Electrical & Computer Engineers:** Supported 120 students in circuit analysis and debugging through weekly labs, office hours, grading, and technical guidance

PROJECTS

12S → 24 V Avionics Power Converter | Altium, Power Electronics, Synchronous Buck, DroneCAN

- Designed, laid out, and validated a 144 W 12S-to-24 V synchronous buck converter achieving 97% efficiency at 6 A, replacing a dedicated avionics battery by powering flight electronics from the aircraft's main flight battery
- Engineered transient protection and input filtering for motor-induced bus disturbances, integrated input/output current sensing, and streamed power telemetry over DroneCAN to the Pixhawk flight controller and ground station

Autonomous Drawing Car | RP2040, C, Pico SDK, Protothreads, UDP | *Circuit Cellar Magazine, Oct. 2026*

- Developed distributed embedded firmware across two Pico W microcontrollers, coordinating wireless commands, IMU sensing, motor actuation, USB input, and real-time system state using C and cooperative Protothreads
- Integrated BNO055 IMU feedback over I²C, PWM motor control, and state-machine navigation to execute wirelessly transmitted waypoints with closed-loop heading control

Floating Box Height Controller | FRDM-KL46Z, C, ToF Sensing, PID, PWM

- Built a closed-loop system using real-time ToF distance sensing, noise filtering, PID control, and PWM servo control to wind and unwind a motorized spool, maintaining constant box height as its support moved vertically
- Debugged incorrect system behavior through systematic subsystem testing and oscilloscope measurements, tracing the fault to incorrect clock scaling in the PIT timing logic and correcting PWM output

SKILLS

Electrical Design: Schematic Capture, PCB Layout, Board Bring-Up, Power Electronics, Power Distribution, Sensor & Actuator Interfaces, Hardware Validation, Root-Cause Debugging, Systems Integration

Hardware & Tools: Altium, Xpediton, LTspice, Oscilloscopes, Logic Analyzers, DMMs, Electronic Loads, Soldering

Embedded: RP2040/RP2350, ATmega328P, Raspberry Pi, Pixhawk, Real-Time Systems, PID Control

Programming Languages: C/C++, Rust, Assembly, Python, Java, Verilog/SystemVerilog

Interfaces & Protocols: I²C, UART, PWM, DroneCAN, UDP